VIVEKANAND EDUCATION SOCIETY'S INSTITUTE OF TECHNOLOGY

(An Autonomous Institute Affiliated to University of Mumbai)

Department of Computer Engineering

Project Report on

MERIDIAN: A Longitudinal Clinical Decision Engine for Prescription Safety

In partial fulfillment of the Third Year (Semester–VI), Bachelor of Technology (B.Tech.) Degree in Computer Engineering at the University of Mumbai

Academic Year 2025-26

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CERTIFICATE of Approval

This is to certify that **Prachi Lund D12A 39, Bhoomi Chhabria D12A 12, Mahek Hingorani D12B 18, Navya Rangwani D12B 43** of Third Year Computer Engineering studying under the University of Mumbai has satisfactorily presented the project on **MERIDIAN: A Longitudinal Clinical Decision Engine for Prescription Safety** as a part of the coursework of CAPSTONE PROJECT - I for Semester-VI under the guidance of **Dr. Machhindranath Patil** in the year 2025-26.

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Date

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Internal Examiner External Examiner

_____ _____ _____

Project Mentor Head of the Department Principal

Dr. Mrs. Nupur Giri Dr. J. M. Nair

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We are deeply indebted to Head of the Computer Department **Dr.(Mrs.) Nupur Giri** and our Principal **Dr. (Mrs.) J.M. Nair ,** for giving us this valuable opportunity to do this project.

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We wish to express our profound thanks to all those who helped us in gathering information about the project. Our families too have provided moral support and encouragement several times.

Computer Engineering Department

COURSE OUTCOMES FOR CAPSTONE PROJECT

Learners will be to:-

Course Outcomes	Description of the Course Outcomes
CO1	Identify and define a real-world problem through systematic requirement analysis and investigation.
CO2	Conduct a thorough literature review and benchmarking to propose innovative solutions.
CO3	Develop a structured project plan with well-defined milestones, timelines, and risk mitigation strategies.
CO4	Design and implement a working prototype using appropriate engineering tools and methods.
CO5	Analyze experimental results, validate the solution, and optimize performance based on data.
CO6	Effectively document and present the project outcomes, demonstrating professional ethics, teamwork, and project management skills.

ABSTRACT

The rapid digitization of healthcare systems has resulted in the generation of large volumes of patient data; however, this data often remains fragmented across multiple healthcare providers, paper-based records, and isolated digital systems. Such fragmentation significantly impacts the quality of clinical decision-making, particularly in prescription practices, where incomplete patient history may lead to adverse drug interactions, redundant diagnostic procedures, and poor continuity of care. These challenges are especially prominent in developing healthcare ecosystems like India.

MERIDIAN: A Longitudinal Clinical Decision Engine for Prescription Safety is proposed as an intelligent system that transforms fragmented medical records into structured, time-oriented clinical intelligence. The system aggregates patient data from multiple sources and organizes it into a comprehensive longitudinal timeline, enabling clinicians to access complete patient history in a single interface.

The core component of MERIDIAN is a decision engine that evaluates prescription safety using multiple parameters, including medication history, laboratory trends, allergies, and comorbid conditions. Based on this analysis, the system generates real-time alerts such as drug-drug interaction warnings, duplicate test detection, overdue diagnostic reminders, and high-risk patient summaries. These features assist clinicians in making informed and safe prescription decisions.

The system is implemented using a MERN (MongoDB, Express.js, React.js, Node.js) stack for the web platform, combined with a Python-based analytical microservice for advanced data processing. MERIDIAN aims to improve prescription safety, reduce healthcare costs, and enable efficient long-term monitoring of chronic diseases. The proposed solution contributes toward building a safer, data-driven, and patient-centric healthcare ecosystem.

MERIDIAN also improves the overall efficiency of healthcare by enabling better data accessibility and reducing the burden on clinicians. It provides clear and quick insights that help in making faster and safer clinical decisions. By supporting organized and intelligent data management, the system enhances patient care and can be easily extended with advanced features in the future.

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CHAPTER 1: INTRODUCTION

This chapter introduces the concept of prescription safety and highlights the challenges associated with fragmented healthcare data. It presents the motivation behind the MERIDIAN system, defines the problem being addressed, and establishes the relevance of the proposed solution in modern healthcare systems.

1.1 Introduction to the Project

Healthcare systems generate vast amounts of patient-related data through clinical visits, laboratory tests, prescriptions, and diagnostic procedures. Despite the availability of such data, the lack of integration across healthcare providers leads to fragmented information storage. In many cases, patient records are distributed across hospitals, clinics, and personal files, making it difficult for clinicians to access complete medical histories during consultations.

MERIDIAN is designed as a longitudinal clinical decision engine that integrates and organizes patient data into a unified timeline. By structuring medical records chronologically, the system enables healthcare providers to track disease progression, treatment responses, and diagnostic patterns over time. This longitudinal perspective plays a critical role in improving clinical decision-making, particularly in prescribing medications.

The system not only consolidates patient information but also applies intelligent analytics to detect potential risks associated with prescriptions. Through its decision engine, MERIDIAN enhances the safety, efficiency, and quality of healthcare delivery.

1.2 Motivation

Prescription errors are a significant concern in healthcare systems worldwide. These errors often arise due to incomplete knowledge of patient history, including prior medications, allergies, and existing health conditions. In countries like India, where patients frequently consult multiple healthcare providers and maintain paper-based records, the risk of such errors is further amplified.

The motivation behind MERIDIAN stems from the need

- Improve patient safety by reducing adverse drug interactions

- Eliminate redundant diagnostic procedures
- Enhance continuity of care across multiple healthcare providers
- Provide clinicians with actionable insights derived from patient data

By addressing these issues, MERIDIAN aims to bridge the gap between data availability and effective clinical utilization.

1.3 Drawbacks of the Existing System

Existing healthcare systems such as Electronic Health Records (EHRs) and Hospital Management Systems (HMS) have improved data storage, but they still face several limitations. One major issue is the fragmentation of patient data, where medical records are scattered across different hospitals and clinics. This makes it difficult for doctors to access complete patient history during consultations.

Another drawback is the lack of a longitudinal view of data. Most systems store information in a visit-based manner rather than organizing it over time, which prevents proper analysis of patient health trends. Additionally, these systems provide limited decision support, meaning they do not actively analyze data to detect drug interactions, allergies, or risks.

There is also a problem of duplicate diagnostic tests, as doctors may unknowingly repeat tests due to unavailability of previous records. Moreover, existing systems lack proactive alert mechanisms, such as reminders for overdue tests or warnings for high-risk prescriptions.

Another limitation of existing systems is the lack of standardization and heavy dependence on manual data entry. Patient data is often stored in different formats, making integration and interpretation difficult. This leads to errors, missing information, and poor coordination among healthcare providers, ultimately affecting the quality of patient care.

These limitations highlight the need for an intelligent and integrated system like MERIDIAN that ensures better prescription safety and continuity of care.

1.4 Problem Definition

The current healthcare ecosystem faces several critical challenges due to fragmented data management systems. These challenges directly impact prescription safety and patient outcomes.

The primary problems identified are:

- Lack of a unified system to integrate patient records from multiple sources
- Incomplete access to patient medical history during clinical decision-making
- Increased risk of prescribing drugs that may interact adversely
- Repetition of diagnostic tests due to absence of prior records
- Inefficient monitoring of chronic diseases over extended periods

The objective of this project is to design and implement a system that consolidates patient data and provides real-time clinical decision support to improve prescription safety and healthcare efficiency.

1.5 Relevance of the Project

The MERIDIAN system is highly relevant in modern healthcare, where access to accurate and complete patient data is essential for safe clinical decision-making. By consolidating patient records from multiple sources and organizing them chronologically, the system allows clinicians to view a patient's medical history clearly, reducing errors caused by incomplete information.

In addition to improving prescription safety, MERIDIAN provides alerts for potential drug interactions, duplicate tests, and overdue reports. These features help clinicians make informed decisions, minimize unnecessary testing, and prevent adverse drug reactions, enhancing both patient safety and the efficiency of healthcare services.

The system also supports longitudinal monitoring of patient health, allowing trends in medications, lab results, and chronic conditions to be tracked over time. This aids in better management of long-term illnesses, ensures consistent follow-up, and promotes continuity of care, demonstrating the value of structured data and simple analytics in improving healthcare outcomes.

1.6 Methodology Used

The MERIDIAN system is developed using a structured approach that focuses on data integration, analysis, and decision support. Initially, patient data is collected from various sources such as prescriptions, lab reports, and medical history, and stored in a centralized database after preprocessing.

The system then organizes this data into a time-ordered clinical timeline, allowing doctors to view patient history in a structured and continuous manner. This helps in understanding disease progression and past treatments.

A decision engine, implemented using a Python-based microservice, analyzes the data by considering factors such as medication history, allergies, lab trends, and comorbidities. Based on this analysis, the system generates alerts like drug interaction warnings, duplicate test detection, and overdue test reminders.

The platform is built using the MERN stack (MongoDB, Express.js, React.js, Node.js), ensuring scalability and efficiency. The development follows an Agile methodology, allowing continuous improvement through testing and feedback.

This methodology ensures that MERIDIAN provides accurate, efficient, and safe clinical decision support.

CHAPTER 2: LITERATURE SURVEY

The literature survey focuses on analyzing existing research work, technologies, and systems related to Clinical Decision Support Systems (CDSS), Electronic Health Records (EHR), and healthcare data analytics. It helps in understanding the strengths and limitations of current systems and identifies the gaps that can be addressed by the proposed system, MERIDIAN. By studying various research papers, books, and expert opinions, a clear understanding of the domain is achieved, which supports the design and development of a more efficient and intelligent healthcare solution.

2.1 Research Papers

Several research papers have been studied to gain insights into the current advancements in healthcare technology and clinical decision support systems.

[1] Advancements in Clinical Decision Support Systems and Digital Health Technologies (2025)

This paper discusses recent advancements in Clinical Decision Support Systems (CDSS) and their integration with digital health technologies such as AI, big data, and cloud computing. It highlights how modern CDSS improves clinical workflows, enhances decision-making, and supports real-time patient monitoring.

Inference:

The study shows that advanced CDSS can significantly improve healthcare quality. It supports MERIDIAN's use of intelligent analytics and real-time alerts for enhancing prescription safety.

[2] Improving Patient Safety Using Health Informatics Systems (2024)

This research focuses on how health informatics systems contribute to patient safety by reducing medical errors and improving data accessibility. It emphasizes the role of integrated data systems in better clinical decision-making.

Inference:

The paper validates that data integration is essential for safe healthcare practices. MERIDIAN addresses this by consolidating fragmented patient records into a unified system.

[3] Longitudinal clinical decision support for assessing decisions over time: State-of-the-art and future directions (2024)

The paper "Longitudinal Clinical Decision Support for Assessing Decisions Over Time" explains that traditional Clinical Decision Support Systems (CDSS) rely on single-time data, which limits accurate clinical decisions. It emphasizes the use of longitudinal patient data, where information such as medical history, lab results, and treatments is analyzed over time. This approach helps in better diagnosis, treatment planning, and monitoring of chronic diseases.

Inference:

The study shows that using time-based data improves the quality and safety of clinical decisions. It supports MERIDIAN's design, which organizes patient records into a chronological timeline and provides better insights for safe and effective prescription decisions.

[4] Agency for Healthcare Research and Quality – Clinical Decision Support Systems (2023)

This source provides an overview of CDSS, including its types, benefits, and role in improving healthcare quality and patient safety. It also explains how CDSS helps clinicians make evidence-based decisions.

Inference:

It establishes the importance of CDSS in modern healthcare, reinforcing the need for systems like MERIDIAN to support safe and informed prescriptions.

[5] Office of the National Coordinator for Health Information Technology – Clinical Decision Support (2023)

This resource describes how CDSS is implemented in healthcare IT systems and its role in improving clinical quality and safety through alerts, reminders, and guidelines.

Inference:

The study highlights the importance of real-time alerts and decision support, which aligns with MERIDIAN's alert-based system for drug interactions and overdue tests.

[6] Towards Effective Clinical Decision Support Systems: A Systematic Review (2022)

This systematic review analyzes multiple CDSS implementations and identifies factors that contribute to their effectiveness, such as usability, accuracy, and integration with workflows.

Inference:

The paper shows that CDSS must be well-designed and user-friendly. MERIDIAN incorporates these aspects by providing structured data and meaningful alerts.

[7] A systematic review of the value of clinical decision support systems in the prescription of antidiabetic drugs (2024)

This paper analyzes the role of Clinical Decision Support Systems (CDSS) in improving prescription practices for patients with diabetes. It focuses on how CDSS helps clinicians manage complex medication regimens by analyzing patient data such as blood glucose levels, medical history, and ongoing treatments. The study shows that CDSS reduces medication errors, improves treatment accuracy, and supports better management of chronic diseases like diabetes.

Inference:

The research highlights that CDSS is highly effective in managing chronic conditions and ensuring safe prescriptions. It supports MERIDIAN's approach of using patient history and data analysis to improve prescription safety and long-term disease monitoring.

[8] Improving Prescription Safety Using Intelligent Healthcare Systems (2024)

This research focuses on intelligent healthcare systems that improve prescription safety by analyzing patient data, detecting drug interactions, and reducing medication errors.

Inference:

The study directly supports MERIDIAN's objective of enhancing prescription safety through intelligent data analysis and decision support.

2.2 Books / Articles / Newspaper Referred

The development of the MERIDIAN system is supported by various books, research articles, and credible publications related to healthcare informatics, clinical decision support systems, and data integration. These references provide the theoretical foundation and practical insights required for designing an efficient and safe prescription system.

Books Referred

- Biomedical Informatics: Computer Applications in Health Care and Biomedicine – This book provides a comprehensive understanding of healthcare data systems, clinical decision support, and medical informatics.
- Health Informatics: Practical Guide for Healthcare and Information Technology Professionals – Offers practical insights into implementation of healthcare IT systems and data management.
- Introduction to Health Care Management – Helps in understanding healthcare workflows, patient data handling, and system management.

Research Articles Referred

- Clinical Decision Support Systems: State of the Art – Discusses the design and importance of decision support systems in improving clinical outcomes.
- The Role of Electronic Health Records in Improving Patient Safety – Explains how EHR systems contribute to safer healthcare practices.
- Medication Errors and Adverse Drug Events in Healthcare Systems – Highlights the causes and impact of prescription errors, reinforcing the need for systems like MERIDIAN.
- Data Integration in Healthcare: Challenges and Opportunities – Provides insights into handling fragmented healthcare data.

Newspapers / Online Sources Referred

- World Health Organization reports and publications – Used for global statistics on medication safety and healthcare challenges.
- Ministry of Health and Family Welfare – Provides India-specific healthcare data and policies.
- The Hindu – Articles related to healthcare digitization and medical infrastructure in India.
- The Times of India – Reports on healthcare challenges, prescription errors, and system improvements.

2.3 Interaction with Domain Experts

Interaction with healthcare professionals and domain experts provided valuable insights into real-world challenges faced in clinical practice. Doctors and practitioners emphasized that one of the major issues in healthcare is the lack of complete patient information during treatment.

Experts explained that incomplete patient history often leads to incorrect prescriptions, missed allergies, and adverse drug reactions. They highlighted the importance of having a unified system that provides complete access to patient records in a structured and easily understandable format.

Additionally, domain experts stressed the need for automated systems that can assist in clinical decision-making by providing alerts, reminders, and risk analysis. Such systems can help reduce the cognitive burden on healthcare professionals and improve the accuracy of diagnosis and treatment.

Experts also suggested that the system should be user-friendly, fast, and reliable, so that it can be easily adopted in real-world clinical environments. Their feedback played a crucial role in defining the features and functionalities of the MERIDIAN system.

2.4 Patent Search

A patent search was conducted to study recent advancements in Clinical Decision Support Systems and intelligent healthcare technologies. The focus was on understanding how modern systems use artificial intelligence, longitudinal data analysis, and real-time clinical insights. Recent developments show a strong trend toward AI-driven, patient-centered, and interoperable decision support systems in healthcare.

Patent 1

Title: Artificial Intelligence System for Predicting Patient Health Events
Patent Number: US20240120109A1
Year: 2024
Link: https://patents.google.com/patent/US20240120109A1/en
Source: Google Patents (USPTO)

Summary:

This patent describes an AI-based system that analyzes patient medical records and historical data to predict future health events. The system uses advanced algorithms to identify patterns in patient data and generate predictive insights. It supports clinicians in early diagnosis and preventive care by providing risk-based recommendations.

Inference:

This system focuses on predictive analytics, while MERIDIAN extends this by integrating prediction with prescription safety alerts and longitudinal timeline analysis.

Patent 2

Title: Clinical Decision Support System for Healthcare Management

Patent Number: EP3456789A1

Year: 2023

Link: http://worldwide.espacenet.com/

Source: Espacenet (European Patent Office)

Summary:

This patent describes a clinical decision support system that processes patient medical records and provides recommendations to healthcare professionals. The system integrates multiple sources of patient data, including medical history, prescriptions, and diagnostic results. It uses rule-based logic to detect potential risks such as incorrect medication or conflicting treatments and assists clinicians in making informed decisions.

Inference:

This system focuses on clinical recommendations, whereas MERIDIAN enhances this approach by incorporating longitudinal timeline analysis and advanced prescription safety checks.

Patent 3

Title: Drug Interaction Detection and Alert System

Patent Number: EP3567891A1

Year: 2024

Link: http://worldwide.espacenet.com/

Source: Espacenet (European Patent Office)

Summary:

This patent describes a system designed to detect drug–drug interactions using patient prescription data. It analyzes combinations of medications and generates alerts when potential adverse interactions are identified. The system improves medication safety and reduces the risk of harmful drug effects.

Inference:

This patent is directly related to MERIDIAN's drug interaction feature. However, MERIDIAN extends this functionality by combining it with lab trend analysis and historical patient data.

Patent 4

Title: Artificial Intelligence-Based Clinical Decision Support System

Patent Number: US20240120109A1

Year: 2024

Link: https://patents.google.com/patent/US20240120109A1/en

Source: Google Patents (USPTO)

Summary:

This patent describes an AI-based system that analyzes longitudinal patient data to predict health risks and support clinical decisions. It uses machine learning algorithms to identify patterns in patient history and generate predictive insights for diagnosis and treatment.

Inference:

This system aligns with MERIDIAN's longitudinal data analysis, but MERIDIAN further includes real-time alerts, duplicate test detection, and prescription safety evaluation.

Patent 5

Title: Electronic Health Record Analysis System for Clinical Decision Support

Patent Number: EP3678902A1

Year: 2022

Link: http://worldwide.espacenet.com/

Source: Espacenet (European Patent Office)

Summary:

This patent focuses on analyzing electronic health records to provide clinical insights. It organizes patient data into structured formats and enables efficient retrieval and analysis. The system supports clinicians by providing relevant medical information for better diagnosis and treatment planning.

Inference:

This system mainly focuses on data organization, while MERIDIAN adds intelligent analysis, longitudinal tracking, and safety alerts.

CHAPTER 3: REQUIREMENTS OF PROPOSED SYSTEM

This chapter outlines the detailed requirements for the MERIDIAN system, including functional and non-functional aspects, constraints, and technologies used. These requirements form the foundation for system design and implementation.

3.1 Functional Requirements

The MERIDIAN system is designed to support efficient handling and analysis of patient data. It must allow healthcare professionals to input and manage detailed patient information, including medical history, prescriptions, laboratory reports, allergies, and existing medical conditions.

A key feature of the system is the creation of a longitudinal clinical timeline, where all patient-related events are arranged chronologically. This enables doctors to easily track disease progression, treatment history, and diagnostic outcomes over time.

The system must also implement a decision engine that evaluates prescription safety. It should analyze multiple factors such as medication history, laboratory trends, allergies, and comorbidities to identify potential risks. Based on this analysis, the system generates alerts for drug interactions, duplicate tests, overdue diagnostics, and high-risk conditions.

Additionally, the system should provide a summarized risk report that highlights critical issues and assists clinicians in making informed decisions quickly.

Decision Engine Analysis

Parameter	Purpose
Medication History	Detect past drugs
Lab Trends	Identify abnormalities
Allergies	Avoid harmful drugs
Comorbidities	Adjust prescriptions
Risk Score	Summarize patient risk

3.2 Non-Functional Requirements

The MERIDIAN system must meet several non-functional requirements to ensure its effectiveness and reliability. Performance is a critical factor, as the system should process data and generate alerts in real time without delays.

Scalability is also important, as the system should be able to handle increasing amounts of patient data as usage grows. Security is a major concern, as healthcare data is sensitive and must be protected using proper authentication and encryption mechanisms.

The system should be highly available and reliable, ensuring that it can be accessed whenever required. Usability is another important factor, and the interface should be designed in a way that healthcare professionals can use it easily without extensive training.

3.3 Constraints

The development and implementation of the MERIDIAN system may face several constraints. One of the major challenges is the availability of complete and accurate patient data, as healthcare records are often fragmented and stored in different formats.

Integration with existing systems can also be difficult due to lack of standardization. Additionally, the system depends on internet connectivity and digital infrastructure, which may not be available in all healthcare settings, especially in rural areas.

Time and resource limitations during development may also affect the scope and complexity of the system.

3.4 Hardware & Software Requirements

The system requires standard computing devices such as desktops or laptops with sufficient processing power and internet connectivity. A web browser is required to access the application interface.

The software stack used for development includes MongoDB for database management, Express.js and Node.js for backend services, and React.js for frontend development. A Python-based microservice is used for implementing the decision engine and performing analytical tasks.

3.5 Techniques Utilized till Date

Several techniques have been applied during the development of the MERIDIAN system. Data preprocessing techniques such as cleaning, normalization, and structuring are used to ensure consistency in patient data.

Another key technique is longitudinal timeline generation, where patient data such as prescriptions, laboratory reports, and medical history are arranged in a chronological order. This enables healthcare professionals to observe patterns, track disease progression, and understand the impact of treatments over time.

The system also uses rule-based analysis techniques to evaluate prescription safety. Predefined medical rules are applied to detect drug interactions, identify contraindications, and flag potential risks based on patient-specific conditions like allergies and comorbidities.

In addition, trend analysis techniques are used to monitor changes in laboratory values and health indicators over time. This helps in identifying abnormal patterns and early signs of health deterioration. Basic analytical methods are also applied to generate summaries and risk scores for patients. These techniques collectively enable the MERIDIAN system to provide accurate, timely, and context-aware decision support to clinicians.

3.6 Tools Utilized till Date

The development process utilizes various tools for coding, database management, and analysis. Visual Studio Code is used as the primary development environment. MongoDB is used for storing and managing patient data.

Backend development is carried out using Node.js and Express.js, while React.js is used to design the user interface. Python libraries are used for implementing the decision engine and performing analytical operations.

3.7 SDGs Mapped with the Proposed Project and Justification

The MERIDIAN system aligns with Sustainable Development Goal 3: Good Health and Well-being, which focuses on improving healthcare quality and reducing mortality rates.

By providing intelligent decision support, MERIDIAN helps in reducing prescription errors and preventing adverse drug interactions. This directly improves patient safety and reduces the risk of complications caused by incorrect medication.The system also helps in minimizing unnecessary diagnostic tests by identifying duplicates and providing access to previous reports. This reduces healthcare costs and avoids unnecessary burden on patients.

Additionally, the system supports long-term monitoring of chronic diseases by maintaining a longitudinal record of patient data. This enables early detection of health issues and promotes preventive care, which is a key aspect of sustainable healthcare. Overall, the MERIDIAN system contributes to improving healthcare accessibility, efficiency, and quality, thereby supporting the broader objectives of sustainable development.

CHAPTER 4: PROPOSED DESIGN

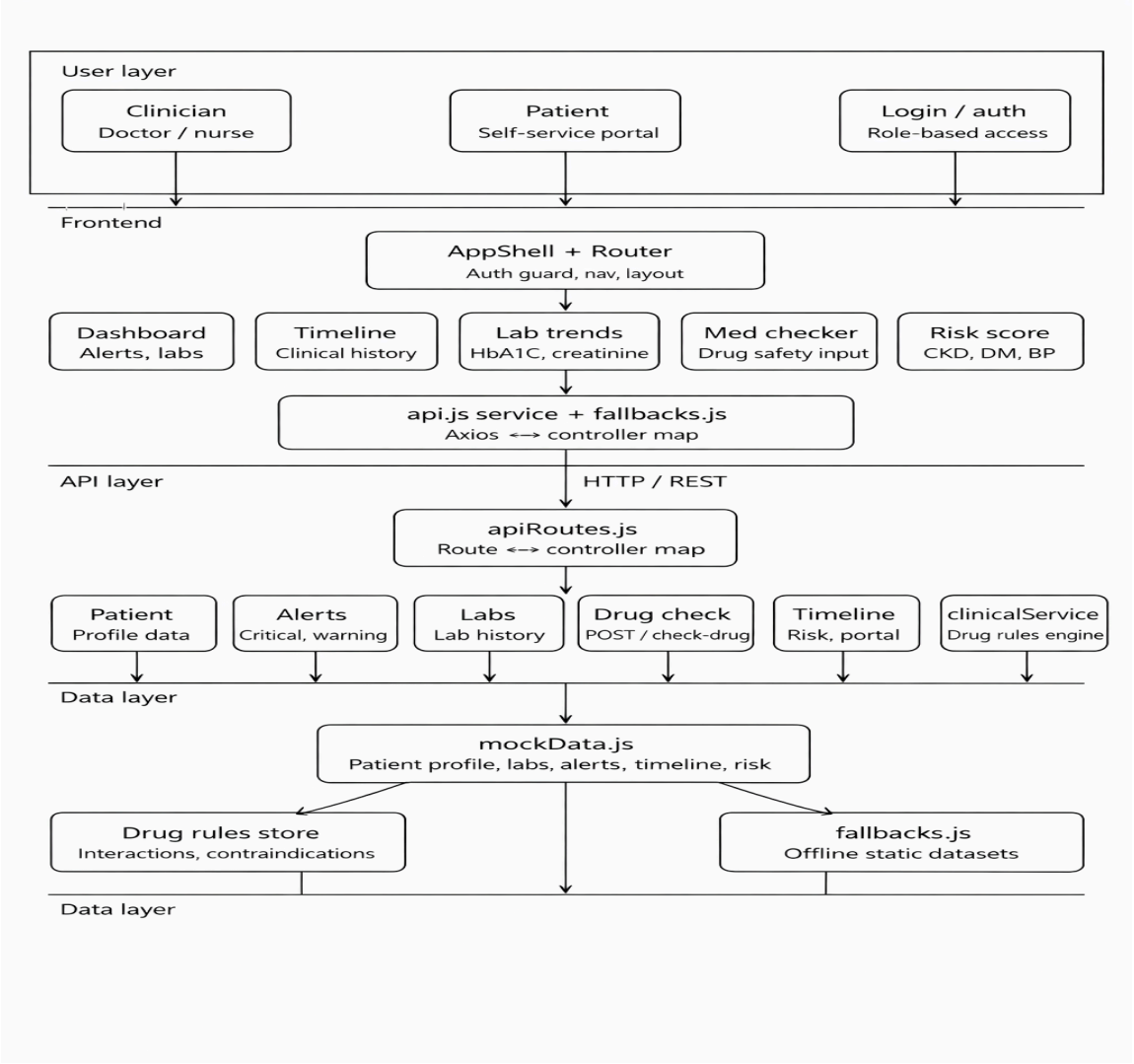
This chapter describes the overall design and architecture of the MERIDIAN system. It explains how different components interact, how data flows through the system, and how the system is structured into modules. It also includes diagrams, algorithms, and project scheduling to provide a complete understanding of the system design.

4.1 Block Diagram Representation

The block diagram of the MERIDIAN system illustrates the high-level architecture and interaction between different system components.

The system consists of the following major blocks:

- **User Interface (Frontend - React.js):**
  Allows clinicians to input patient data, upload reports, and view alerts and timelines.
- **Backend Server (Node.js & Express.js):**
  Handles API requests, manages communication, and processes incoming data.
- **Database (MongoDB):**
  Stores structured patient records including prescriptions, lab reports, allergies, and medical history.
- **OCR Module:**
  Extracts text from uploaded lab reports (PDF/images) and converts it into structured data.
- **Data Preprocessing Unit:**
  Cleans, normalizes, and organizes extracted data into a usable format.
- **Longitudinal Data Builder:**
  Arranges patient data chronologically to create a time-based medical history.



4.2 Modular Diagram Representation of the proposed system with explanation

The modular diagram divides the MERIDIAN system into smaller functional units for better understanding and scalability.

Modules:

Data Input Module

- Accepts patient details, prescriptions, and reports
- Supports structured and unstructured inputs

OCR Processing Module

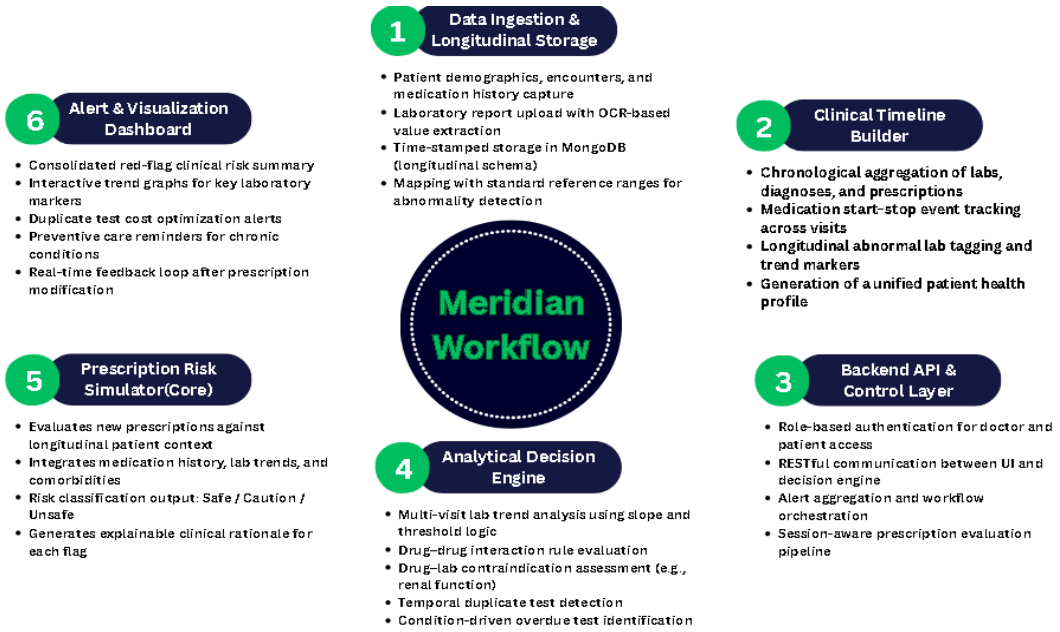
- Extracts text from uploaded lab reports
- Converts unstructured data into machine-readable format

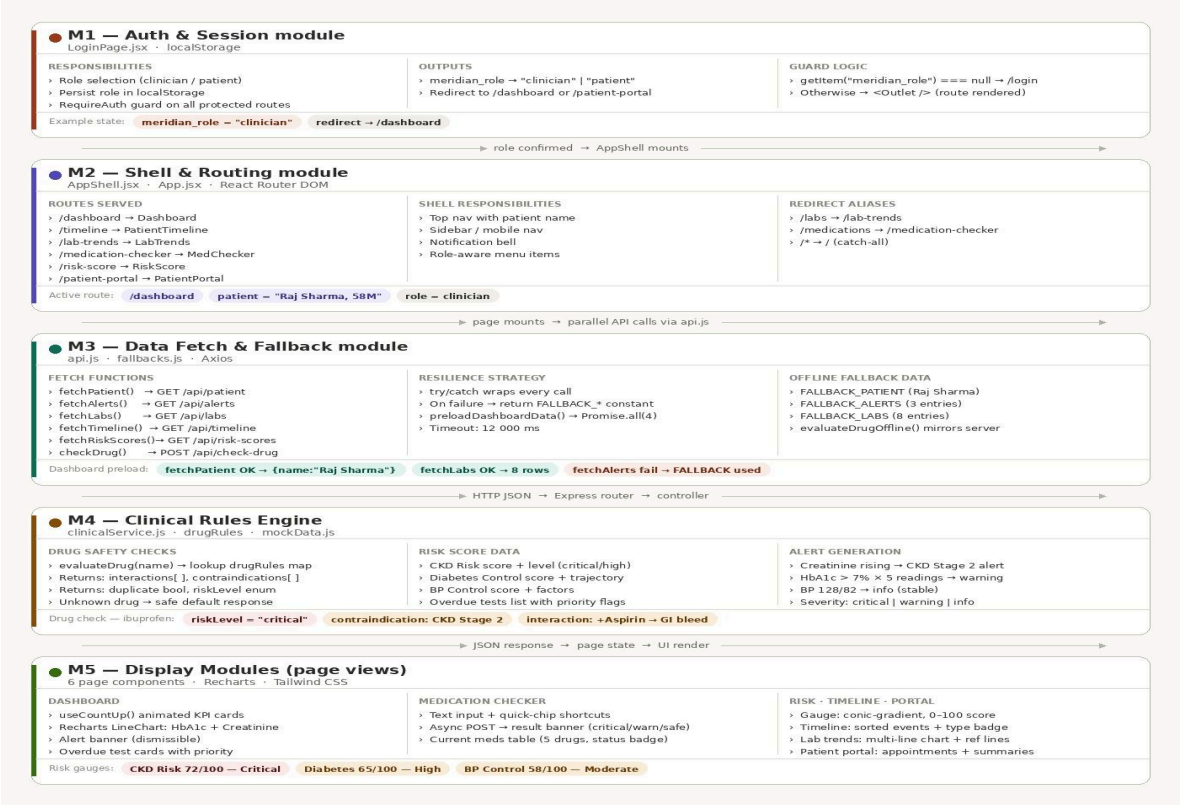
Timeline Generation Module

- Converts data into a chronological sequence
- Enables tracking of disease progression

Decision Engine Module

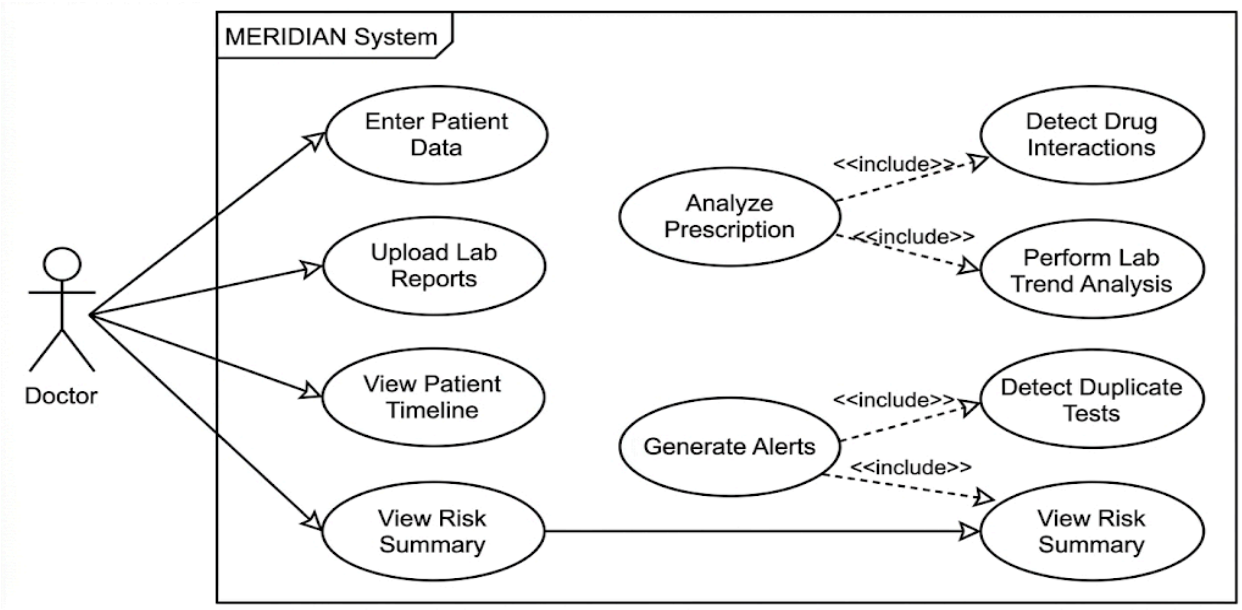
- Core analytical component
- Performs:
    - Drug interaction checks
    - Lab trend analysis
    - Risk scoring



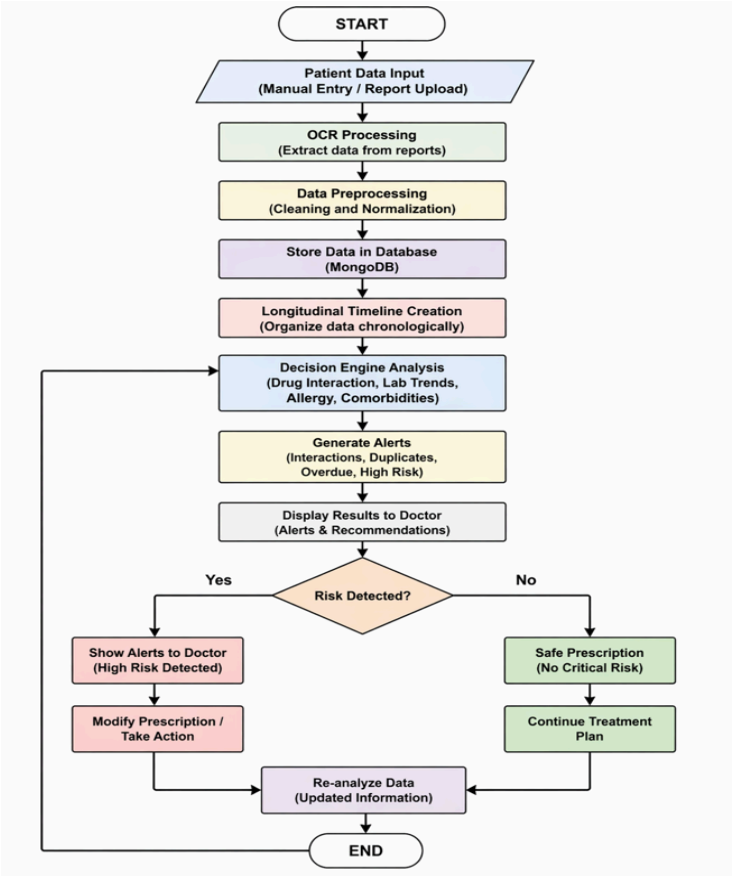


4.3 Design of the proposed system

UML Use Case Diagram for the MERIDIAN Clinical Decision Engine



Flowchart of the Proposed System



The flowchart represents the overall working of the MERIDIAN system. The process begins with the input of patient data, which can be entered manually by the clinician or uploaded in the form of reports. These reports are processed using an OCR module to extract relevant medical information.

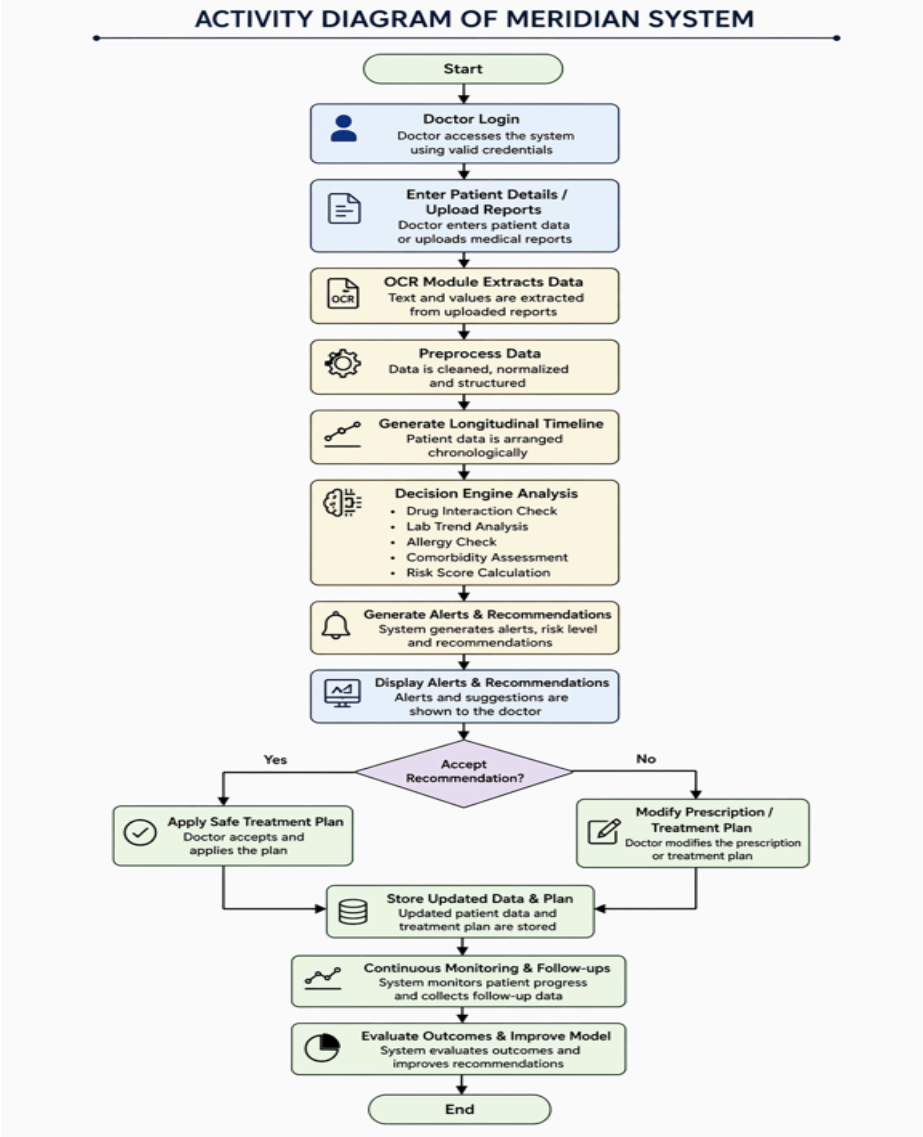
The extracted data is then passed through a preprocessing stage where it is cleaned, normalized, and structured for further analysis. This structured data is stored in a MongoDB database to ensure efficient storage and retrieval.

Next, the system generates a longitudinal timeline by organizing patient data chronologically. This allows clinicians to view the patient's medical history in a continuous and structured manner.

The decision engine then analyzes the data by checking medication history, laboratory trends, allergies, and comorbid conditions. Based on this analysis, the system generates alerts such as drug interactions, duplicate tests, and overdue diagnostics.

The results are displayed to the doctor, who can then review the alerts. If a risk is detected, the prescription is modified and re-evaluated, forming a feedback loop. If no risk is found, the process continues safely. This iterative loop ensures accurate and safe clinical decision-making.

Activity Diagram of the proposed System



The activity diagram represents the workflow and dynamic behavior of the MERIDIAN system in a structured and sequential manner. It illustrates how patient data is processed, analyzed, and used for clinical decision-making.

The process begins with the Start node, followed by the doctor logging into the system using valid credentials. After successful login, the doctor enters patient details or uploads medical reports.

The system then activates the OCR (Optical Character Recognition) module, which extracts relevant textual and numerical data from the uploaded reports. This extracted data undergoes preprocessing, where it is cleaned, normalized, and converted into a structured format suitable for analysis.

Next, the system generates a longitudinal timeline, organizing the patient's medical history chronologically. This enables better understanding of past treatments, lab results, and disease progression.

The core component, the Decision Engine, performs multiple analyses including:

- Drug interaction checking
- Laboratory trend analysis
- Allergy verification
- Comorbidity assessment
- Risk score calculation

Based on this analysis, the system generates alerts and recommendations, which are displayed to the doctor for review. A decision point is then reached where the doctor decides whether to accept the recommendation:

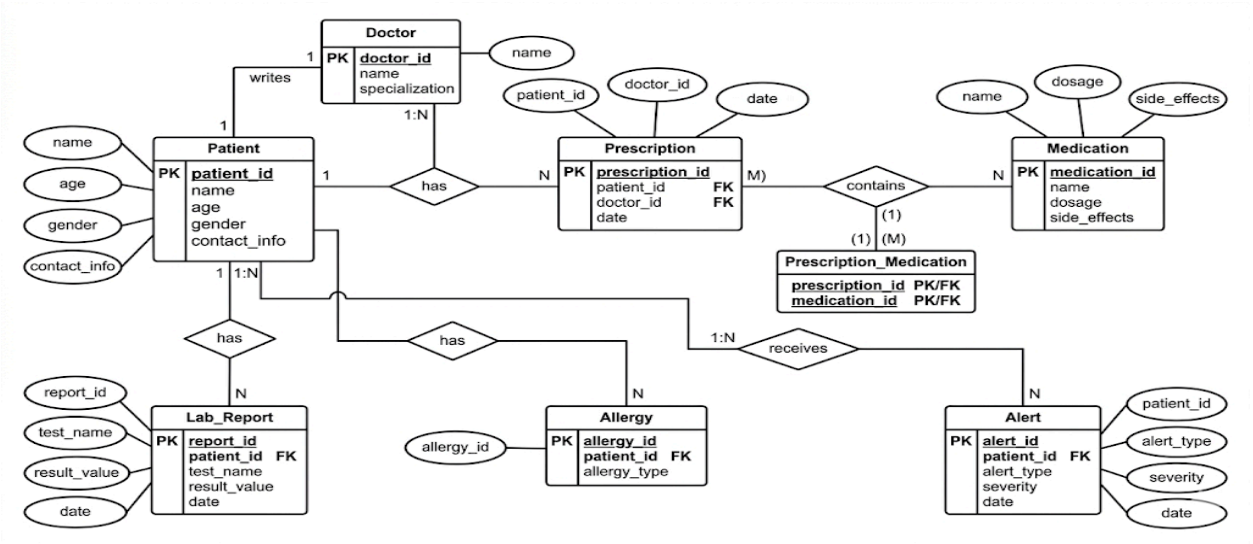
- If accepted, a safe treatment plan is applied
- If rejected, the doctor modifies the prescription or treatment plan

In both cases, the updated data is stored in the system. The system then performs continuous monitoring and follow-ups, tracking patient progress over time. Finally, it evaluates outcomes and refines its decision-making process, leading to improved recommendations in future cycles. The process ends at the End node, completing one full cycle of intelligent clinical decision support.

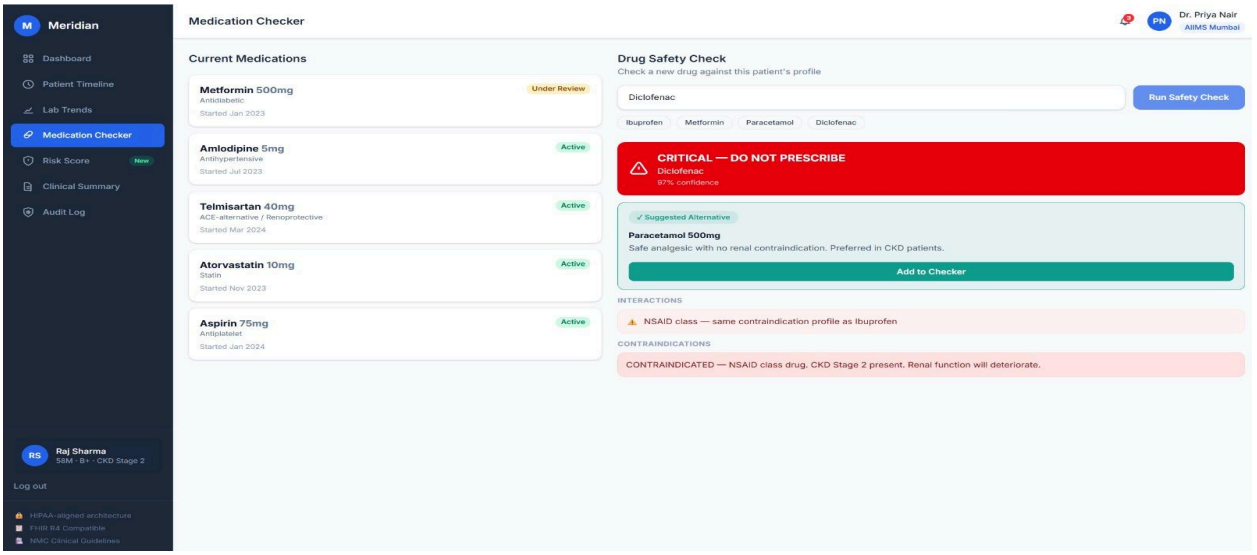
Entity Relationship Diagram for the MERIDIAN Clinical Decision Engine

Architectural Features:

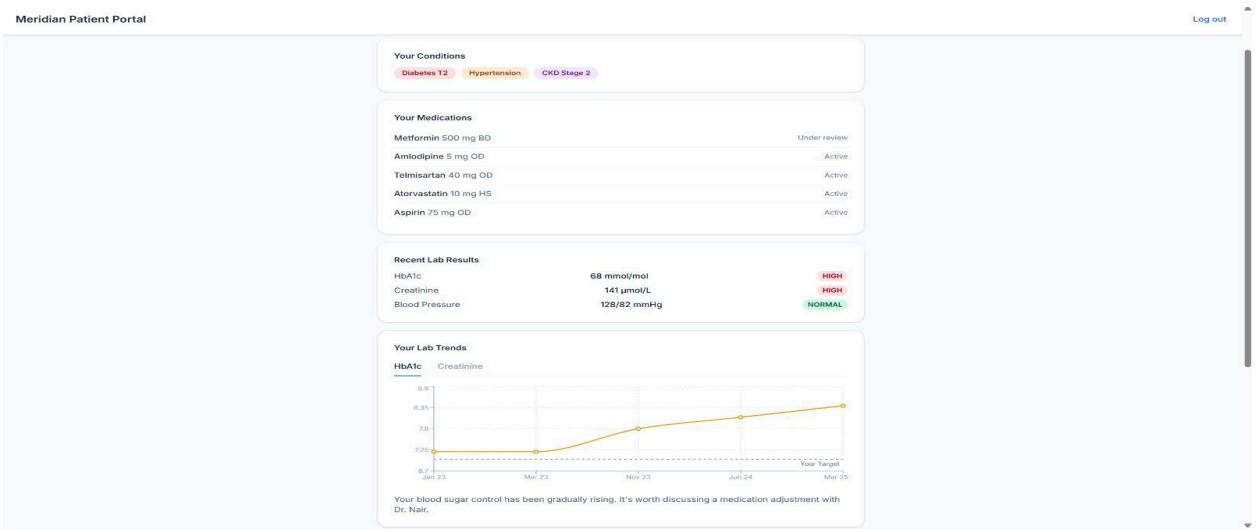
- **Normalization:** Created the Prescription_Medication associative entity to handle the M:N relationship, ensuring the database adheres to 3rd Normal Form (3NF).
- **Primary & Foreign Keys:** All primary keys (PK) are underlined, and foreign keys (FK) clearly show the relational links between patient data, doctor records, and clinical outputs.
- **Clinical Logic:** Longitudinal Tracking: The Lab_Report and Allergy entities are linked to the Patient to allow the engine to query historical data.
- **Safety Layer:** The Alert entity serves as the output of the decision engine, mapped directly to specific patients with severity levels.



Screenshot of Implementation



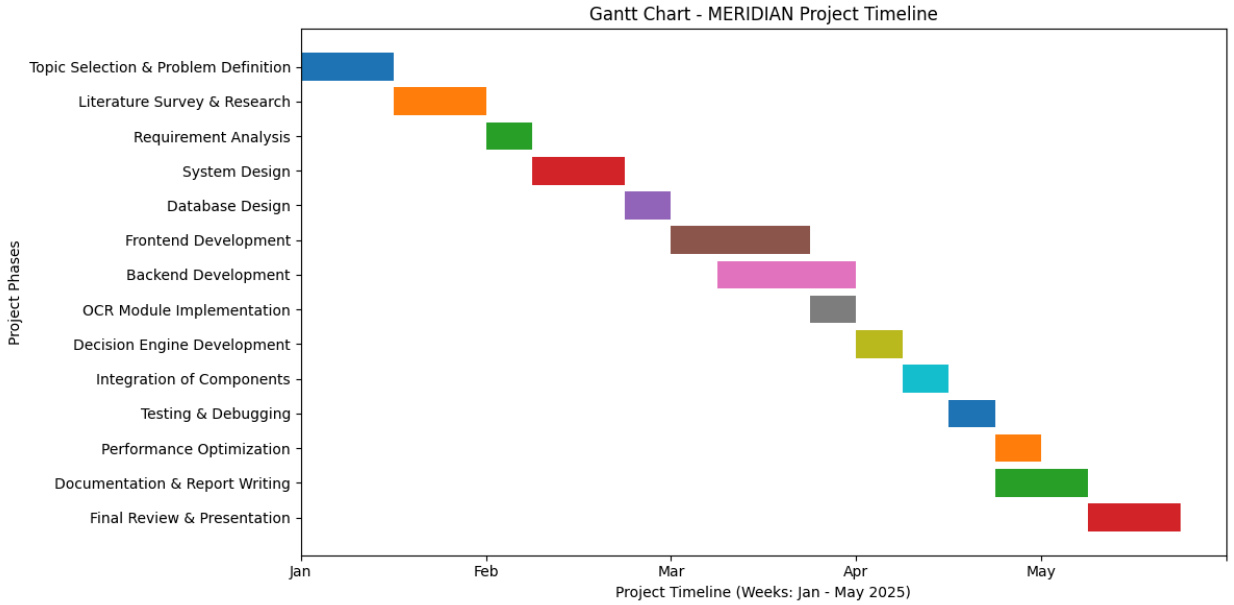




4.4 Algorithms Utilized in Existing Systems

Sr. No.	Technique / Algorithm	Application in Healthcare Systems	Key Limitations
1	Rule-Based Systems	Used in basic Clinical Decision Support Systems (CDSS) for generating alerts based on predefined rules	Rigid, not adaptable to complex or dynamic medical scenarios
2	Machine Learning Models	Predict disease risk, patient outcomes, and assist in diagnosis	Requires large labeled datasets and lacks interpretability
3	NaturalLanguage Processing (NLP)	Extracts information from unstructured clinical notes and reports	Prone to errors due to ambiguity in medical text
4	Electronic Health Record (EHR) Systems	Store and manage patient medical history and reports	Data is often fragmented and lacks integrated analysis
5	StatisticalAnalysis Methods	Analyze trends in lab reports and patient history	Limited capability in capturing complex relationships
6	Traditional CDSS	Provide recommendations and alerts for prescriptions	Mostly static, lacks real-time intelligence adaptability
7	Data Mining Techniques	Identify patterns and correlations in healthcare data	Time-consuming and not Suitable for real-time decision-making

4.5 Project Schedule: MERIDIAN - Gantt Chart



CHAPTER 5: IMPLEMENTATION, RESULTS AND DISCUSSION

This chapter presents the detailed implementation of the MERIDIAN system and discusses the results obtained from its evaluation. It explains how fragmented patient data is collected, processed, and transformed into structured longitudinal information, which is then analyzed by the clinical decision engine. The chapter also evaluates the effectiveness of the system in improving prescription safety and highlights how it overcomes the limitations of existing healthcare systems.

5.1 Data Collection / Acquisition

The MERIDIAN system is designed to address the issue of fragmented healthcare data by enabling centralized data acquisition through a unified platform. In traditional healthcare settings, patient information is distributed across paper-based records and multiple providers, leading to incomplete clinical understanding. To overcome this limitation, the system allows systematic collection and integration of patient data.

The data collected includes patient demographic details such as age, gender, and identification information, along with clinical attributes such as diagnoses, medication history, allergies, and laboratory test results. This information is essential for performing accurate clinical analysis and ensuring safe prescription practices.

Data acquisition is performed through two primary mechanisms. First, structured data entry is carried out through a web-based interface, where clinicians input patient details and prescription information. Second, unstructured laboratory reports are uploaded in the form of PDF documents or images. These reports contain valuable diagnostic information that is not directly usable in raw form.

To ensure robustness and proper validation of the system, synthetic datasets are also incorporated. These datasets simulate real-world chronic disease conditions such as diabetes and hypertension, allowing the system to be tested under diverse clinical scenarios. This combination of structured input and simulated data ensures comprehensive data coverage and reliable system performance.

5.2 Data Preprocessing Techniques

The raw data collected from multiple sources is often inconsistent, incomplete, and unstructured. Therefore, preprocessing plays a critical role in preparing the data for accurate analysis by the decision engine.

The first stage of preprocessing involves extracting textual data from uploaded laboratory reports using an OCR (Optical Character Recognition) engine. Since OCR output may contain noise and formatting inconsistencies, it is further processed to remove errors and irrelevant content.

The next step is data cleaning, where missing values are handled, incorrect entries are corrected, and redundant information is removed. This ensures that the dataset maintains high quality and reliability.

Following cleaning, normalization is performed to standardize units of measurement across different reports. For example, laboratory values expressed in different units are converted into a consistent format to enable meaningful comparison.

The system then structures the extracted data into predefined formats, organizing it into attributes such as test name, measured value, and reference range. Each laboratory value is compared against clinically accepted thresholds to identify abnormal results, which are automatically flagged for further evaluation.

Finally, the longitudinal data builder arranges all patient information in chronological order. This step is essential for enabling time-series analysis, allowing the system to track disease progression and detect trends over time.

5.3 Experimental Setup

The MERIDIAN system is implemented using a hybrid architecture that combines web technologies with a dedicated analytical engine. The frontend is developed using React.js, providing a responsive and interactive interface for clinicians. The backend is implemented using Node.js and Express.js, which manage data processing, API handling, and communication between modules.

MongoDB is used as the database for storing patient records due to its flexibility in handling semi-structured and heterogeneous data. It supports efficient storage and retrieval of longitudinal patient information.

The analytical component of the system is implemented using a Python-based microservice. This module performs advanced data analysis, including laboratory trend evaluation, drug interaction detection, and prescription risk simulation. Communication between the backend and the analytical engine is achieved through RESTful APIs, ensuring modularity and scalability.

The system is deployed and tested on a standard computing environment with adequate processing capability and internet connectivity. Various test scenarios are created to evaluate system performance, including cases involving multiple medications, abnormal laboratory trends, and chronic disease conditions. This experimental setup ensures realistic testing and validates the system's practical applicability.

5.4 Evaluation Metrics

The performance of the MERIDIAN system is evaluated using a set of well-defined metrics that measure both accuracy and efficiency.

One of the primary metrics is the accuracy of laboratory trend detection, which evaluates the system's ability to correctly identify patterns and abnormalities in patient health over time. Another critical metric is the correctness of drug interaction and contraindication alerts, which ensures that potentially harmful medication combinations are accurately detected and reported.

The system also measures duplicate test detection accuracy, which helps in identifying unnecessary repeated diagnostic procedures and contributes to cost reduction. Response time is evaluated to ensure that the system generates alerts and recommendations promptly, supporting real-time clinical decision-making.

Usability is another important factor, which assesses how easily clinicians can interact with the system and interpret the generated insights. Additionally, the effectiveness of red-flag risk summaries is analyzed to determine how clearly and accurately critical risks are communicated to the user.

5.5 Gap Analysis of Existing Systems

Existing healthcare information systems primarily function as repositories for storing patient data and lack advanced analytical capabilities. They do not maintain a structured longitudinal view of patient history, which is essential for understanding disease progression and making informed clinical decisions.

A major limitation of these systems is their inability to provide real-time decision support. Clinicians often rely on isolated and current data without access to complete historical information, increasing the risk of adverse drug interactions and delayed diagnosis. Furthermore, existing systems do not effectively detect duplicate tests or provide reminders for overdue investigations, leading to inefficiencies and increased healthcare costs.

The MERIDIAN system addresses these limitations by transforming fragmented patient records into a structured longitudinal timeline. It integrates data from multiple sources and applies intelligent analysis through a clinical decision engine. The system provides real-time alerts for drug interactions, laboratory abnormalities, duplicate tests, and overdue investigations.

By combining data organization, analytical processing, and visualization in a unified platform, MERIDIAN enhances prescription safety, improves continuity of care, and supports evidence-based clinical decision-making. This makes it a significant improvement over traditional healthcare systems.

Existing vs Proposed System

Feature	Existing Systems (EHR/HMS)	MERIDIAN System
Data Storage	Fragmented	Unified
Data View	Visit-based	Longitudinal timeline

Decision Support	Limited	Intelligent alerts
Drug Interaction	Not reliable	Advanced detection
Duplicate Tests	Not detected	Automatically detected
Alerts	Minimal	Real-time alerts
Data Analysis	Basic	Analytical + rule-based

CHAPTER 6: PLAN OF ACTION FOR THE NEXT SEMESTER

This chapter outlines the progress achieved in the current phase of the MERIDIAN project and presents the planned activities for the next phase. It provides a clear understanding of the development status and the steps required to complete the system successfully. The focus is on enhancing system capabilities, improving performance, and ensuring full implementation of the proposed clinical decision engine.

6.1 Work Done Till Date

In the initial phase of the MERIDIAN project, significant progress has been made in understanding the problem domain and developing the foundational components of the system. The project began with the selection of the topic, followed by an in-depth study of healthcare challenges related to fragmented patient records and prescription safety. A comprehensive literature survey was conducted to analyze existing clinical decision support systems and identify their limitations.

Based on the insights gained, the system architecture was designed, including the block diagram and modular structure. The design focused on creating a scalable and modular system capable of handling longitudinal patient data and supporting real-time analysis.

The development phase included the implementation of the frontend interface using React.js, which allows users to input patient data, upload reports, and view results. The backend was developed using Node.js and Express.js to manage data processing and communication between different modules. MongoDB was integrated as the database to store patient records in a flexible and structured format.

The OCR module was implemented to extract data from laboratory reports, converting unstructured information into machine-readable format. Additionally, the initial version of the clinical decision engine was developed using Python, incorporating basic functionalities such as lab trend analysis, drug interaction detection, and alert generation.

Preliminary testing was carried out using sample and synthetic datasets to verify the working of the system. These activities have established a strong foundation for further development and refinement of the MERIDIAN system.

Work Done Summary Table

Sr. No.	Task	Description
1	Problem Formulation	Identified issues in fragmented healthcare data and unsafe prescriptions
2	Literature Survey	Studied EHR systems, CDSS, OCR, and clinical decision support techniques
3	Data Collection	Collected patient data and medical reports for processing
4	OCR Integration	Extracted text from medical reports using OCR module
5	Data Preprocessing	Cleaned, normalized, and structured patient data
6	Timeline Generation	Created longitudinal patient history timeline
7	Decision Engine Design	Developed rule-based system for risk analysis
8	Alert System	Implemented alerts for drug interactions, duplicate tests, and risks
9	System Design	Created UML diagrams (Use Case, Flowchart, Activity Diagram,

		ER Diagram)
10	Prototype Development	Built frontend (React) and backend (Node.js) system
11	Database Integration	Used MongoDB for storing structured patient data
12	Testing & Evaluation	Tested system functionality and performance
13	Dashboard Development	Developed interface for visualization and monitoring

6.2 Plan of Action for Project II

In the next phase of the MERIDIAN project, the primary focus will be on enhancing the functionality, accuracy, and robustness of the system to make it a complete and reliable clinical decision support platform. The current implementation provides a foundational structure; however, further improvements are required to achieve a production-level system.

One of the key objectives will be to strengthen the clinical decision engine by incorporating a more comprehensive set of medical rules and expanding the drug interaction and contraindication datasets. This will improve the accuracy of risk detection and ensure that the system can handle complex prescription scenarios involving multiple medications, comorbidities, and abnormal laboratory values. Additionally, more advanced logic will be implemented for lab trend analysis, enabling early detection of disease progression in chronic conditions such as diabetes and hypertension.

Another important area of development will be the enhancement of the longitudinal data model. The system will be improved to better represent patient history over time, allowing deeper analysis of clinical patterns and more accurate prediction of potential risks. Integration of additional data sources, such as detailed diagnostic reports and extended medical histories, will further strengthen the analytical capabilities of the system.

The user interface and visualization components will also be refined to improve usability and clarity. Advanced dashboards will be developed to display patient timelines, risk summaries, and trend graphs in a more interactive and intuitive manner. Special attention will be given to designing clear and meaningful alert messages so that clinicians can quickly understand the severity and context of each risk. The real-time alert mechanism will be further optimized to ensure faster response and efficient processing. Alerts such as drug interactions, duplicate tests, and overdue investigations will be generated with improved accuracy and presented with proper prioritization levels. This will help clinicians make quick and informed decisions during patient care.

Extensive testing and validation will be conducted using larger and more diverse datasets to evaluate the performance of the system under different clinical scenarios. This will include edge cases such as patients with multiple chronic conditions and complex medication histories. Based on testing results, necessary improvements will be made to enhance system reliability.

Performance optimization will also be carried out to improve system speed, scalability, and efficiency. The architecture will be refined to handle higher data loads and support multiple users simultaneously. In addition, basic security measures, such as role-based access control and secure data handling, will be implemented to protect sensitive patient information.

Finally, the project will be completed with proper documentation, report finalization, and preparation of presentation materials. The aim of this phase is to transform MERIDIAN into a fully functional, user-friendly, and scalable clinical decision support system that can effectively assist healthcare professionals in improving prescription safety and patient care.

CHAPTER 7: CONCLUSION

The MERIDIAN project addresses a critical challenge in modern healthcare: the fragmentation of patient data and its impact on prescription safety. In real-world clinical settings, patient information is often distributed across multiple providers, paper records, and disconnected digital systems. This fragmentation prevents clinicians from accessing a complete and accurate medical history. As a result, healthcare providers face increased risks of prescription errors, adverse drug interactions, redundant diagnostic tests, and overall inefficiencies in patient care.

MERIDIAN tackles this issue by introducing a longitudinal clinical decision support system that consolidates patient data into a unified, structured, and time-oriented format. By presenting a comprehensive view of a patient's medical history over time, the system enables more informed decision-making, improves prescription accuracy, and enhances the quality and safety of care

The system is designed to transform raw and unstructured healthcare data into meaningful clinical insights. By integrating data collection, preprocessing, and analysis within a unified platform, MERIDIAN provides a comprehensive view of patient history. The longitudinal timeline enables clinicians to track disease progression, monitor treatment outcomes, and identify patterns over time. This structured representation significantly improves the quality of clinical decision-making, particularly in complex cases involving chronic diseases and multiple medications.

A key strength of the MERIDIAN system lies in its intelligent decision engine, which evaluates prescription safety using multiple parameters such as medication history, laboratory trends, allergies, and comorbid conditions. Based on this analysis, the system generates real-time alerts, including drug–drug interaction warnings, duplicate test detection, overdue diagnostic reminders, and high-risk patient summaries. These features not only enhance patient safety but also reduce unnecessary medical procedures and associated healthcare costs. The system thus supports clinicians in making faster, more accurate, and evidence-based decisions.

From a technical perspective, the implementation of MERIDIAN demonstrates the effective use of modern technologies such as the MERN stack for web development and Python-based

microservices for analytical processing. The modular architecture ensures scalability, flexibility, and efficient handling of diverse healthcare data. Additionally, the incorporation of OCR-based data extraction and preprocessing techniques enables the system to handle both structured and unstructured data sources, making it practical for real-world deployment.

Another significant contribution of the system is its ability to improve continuity of care. By maintaining a comprehensive and time-ordered record of patient data, MERIDIAN facilitates long-term monitoring of health conditions and supports better coordination among healthcare providers. This is particularly beneficial in managing chronic diseases, where continuous tracking and timely intervention are essential for effective treatment.

In summary, MERIDIAN serves as an effective approach to improving prescription safety and the quality of healthcare delivery. It addresses the gap between fragmented data and clinical decision-making through structured data integration and intelligent insights. Looking ahead, the inclusion of machine learning capabilities, enhanced system interoperability, and broader clinical data integration can further strengthen its role as a comprehensive, data-driven clinical decision support system.

CHAPTER 8: REFERENCES

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